

Titanic_Logistic_Regression.py

```
import pandas as pd
from sklearn.linear_model import LogisticRegression

# Load Titanic dataset
data = pd.read_csv("Titanic/train.csv")

# Select features
X = data[["Pclass", "Age", "Fare"]]

# Result: 0 = Not Survived, 1 = Survived
y = data["Survived"]

# Handle missing values
X = X.fillna(X.mean())

# Create model
model = LogisticRegression()

# Train model
model.fit(X, y)

# Predict for a passenger
new_passenger = pd.DataFrame(
    [[3, 25, 10]],
    columns=["Pclass", "Age", "Fare"]
)

prediction = model.predict(new_passenger)

print("Prediction:", prediction)

# Probability
probability = model.predict_proba(new_passenger)

print("Probability:", probability)
```

Output:

```
• (venv) PS C:\Users\Jites\Desktop\New folder> & "c:\Users\Jites\Desktop\New folder\venv\Scripts\python.exe" "c:/Users/Jites/Desktop/New folder/Titanic/Titanic_Logistic_Regression.py"
Prediction: [0]
Probability: [[0.74954176 0.25045824]]
```

Titanic_Logistic_Regression_Train_Test_Accuracy.py

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Load Titanic dataset
data = pd.read_csv("Titanic/train.csv")
```

```
# Select features
X = data[["Pclass", "Age", "Fare"]]

# Result: 0 = Not Survived, 1 = Survived
y = data["Survived"]

# Handle missing values
X = X.fillna(X.mean())

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create model
model = LogisticRegression()

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

# New passenger
new_passenger = pd.DataFrame(
    [[3, 25, 10]],
    columns=["Pclass", "Age", "Fare"]
)

prediction = model.predict(new_passenger)
probability = model.predict_proba(new_passenger)

print("Prediction:", prediction)
print("Probability:", probability)
```

Output:

```
(venv) PS C:\Users\Jites\Desktop\New folder> & "c:\Users\Jites\Desktop\New folder\venv\Scripts\python.exe" "c:/Users/Jites/Desktop/New folder/Titanic/Titanic_Logistic_Regression_Train_Test_Accuracy.py"
Accuracy: 0.7374301675977654
Prediction: [0]
Probability: [[0.74626065 0.25373935]]
```